

# Chandan Abhishek Muchukota

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## Summary

Software Engineer specializing in scalable distributed systems and cloud-native infrastructure. Skilled in building event-driven architectures and data pipelines that process millions of events daily, with experience integrating LLM-powered services into production workflows. Strong foundation in system design, performance optimization, and enhancing reliability for large-scale applications.

## Accomplishments

- Served as a **founding engineer** at multiple early-stage startups (CodeGreenBack, Phamoss, CISApp), leading development of scalable web and mobile platforms.
- Engineered the **fastest online compiler** for the CodeGreenBack platform using **Go (Golang)** and developed large-scale backend systems including **production APIs serving 1M+ users** and **distributed pipelines processing 10M+ events/day**.
- Solved **500+ LeetCode problems**, achieving a 3-star rating and demonstrating strong algorithmic problem-solving ability.

## Experience

### University of Florida

Jan 2025 - Present

#### Software Engineer

Gainesville, FL

- Architected a geospatial digital-twin platform for resilience planning using **React, TypeScript, ArcGIS JS SDK, Dash, and Plotly**, enabling interactive analysis across 250K+ geospatial building records.
- Designed scalable geospatial APIs leveraging **FastAPI, SQLAlchemy, and PostGIS** and created comprehensive documentation and onboarding materials, reducing new developer ramp-up time by **40%**.
- Constructed spatial analytics pipelines processing **250K+ building records** and **1M+ metadata attributes**, uncovering flood, heat, and infrastructure vulnerabilities that informed urban planning decisions and scenario-based resilience simulations.
- Orchestrated containerized ETL pipelines using **Apache Airflow on GCP** and Docker, processing **10+ GB geospatial datasets** and loading curated data into BigQuery, improving analytical query performance by **30%**.
- Designed and developed a **multi-agent** conversational assistant, where specialized agents handled geospatial queries, resilience scenario simulations, and analytical workflow execution, coordinated through an **LLM-based orchestration layer** with **tool-calling capabilities** to interface with backend geospatial services - directly applying **machine learning** and **information retrieval** to Geo data workflows.

### PruTech (Client: NYC Comptroller's Office)

May 2025 - Dec 2025

#### Software Engineer Intern

Remote, USA

- Engineered secure microservices-based REST APIs using **C#, .NET Core, and SQL Server** supporting financial systems processing **\$2B+ annually**.
- Optimized high-volume **SQL Server** workloads through indexing and query tuning improving performance by **40%**.
- Built interactive dashboards using **Angular, TypeScript**, and integrated .NET APIs, reducing report generation time by **25%**.
- Automated testing and CI/CD pipelines across the **software development lifecycle (SDLC)** using **NUnit and Azure DevOps**, reducing production defects by **25%**.
- Migrated and secured **2TB+ financial datasets** ensuring compliance with enterprise security standards.

### Infinite Analytics

Dec 2023 - Jul 2024

#### Software Engineer

Mumbai, India

- Delivered backend services using **FastAPI** on AWS powering production recommendation APIs for **1M+ users** in large-scale production systems.
- Architected distributed event-processing pipelines handling 10M+ events/day using message queues with sub-**200ms latency** and **99.99% uptime**.
- Integrated **LLM-powered code review automation** into CI/CD workflows improving development productivity by **30%**.
- Designed event-driven ingestion pipelines using **Kafka and RabbitMQ** reducing processing latency to **under 5 seconds**.
- Designed scalable inference pipelines for **LLM-powered services**, optimizing latency and throughput for production workloads.

### Cogoport

Feb 2023 - Oct 2023

#### Software Engineer

Mumbai, India

- Built backend microservices using **FastAPI and Ruby on Rails** deployed on AWS powering recommendation, forecasting, and analytics APIs serving **1M+ monthly users**.
- Developed a custom **demand forecasting system** for container shipping logistics improving forecast reliability despite noisy operational data.
- Formulated a mathematical forecasting model incorporating **seasonality and trend analysis** achieving **78% prediction accuracy**.
- Reduced excess inventory by **20%** and shipment cancellations by **37%** improving logistics planning and capacity utilization.

## Projects

### Delphi Pascal Interpreter

- Implemented a custom Delphi Pascal interpreter in Java using ANTLR to support object-oriented language features and packaged it as a Docker container for consistent deployment.
- Developed parsing, AST traversal, execution engine, and unit tests using the Visitor design pattern, integrating automated tests into a CI pipeline.
- Improved interpreter performance by **25%** using compile-time evaluation and memory optimization.

## Technical Skills

- Languages:** Python, C#, Java, JavaScript, Go, SQL, TypeScript
- Frameworks:** FastAPI, Ruby on Rails, .NET Core, React, Angular, Node.js, Flutter
- Cloud / Infrastructure:** AWS, Azure, Docker, Apache Airflow, BigQuery
- Databases:** PostgreSQL, MySQL, Cassandra, SQL Server, PostGIS
- Messaging / Streaming:** Kafka, RabbitMQ
- Operating Systems:** Linux, Unix
- Concepts:** Distributed Systems, Microservices, System Design, REST APIs, Data Structures and Algorithms, CI/CD, Scalability, Performance Optimization

## Education

University of Florida | *Master of Science, Computer Science* (GPA: 3.96)

Gainesville, FL Aug 2024 - May 2026